

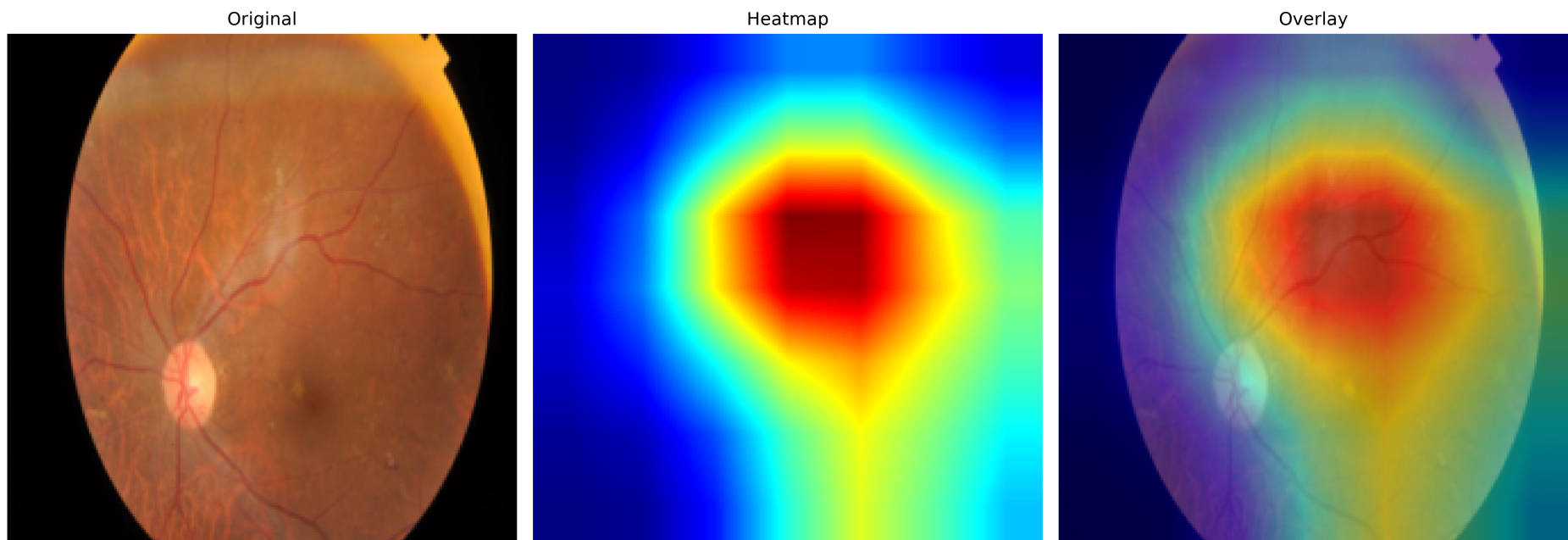
FusionMedAI: XAI Summary Report

Run Details & Metadata

Total images evaluated	367
Correct predictions	309
Incorrect predictions	58
Average confidence	0.9107752442359924
Average entropy	0.2334100753068924
Average inference time (ms)	31.347409905789657
Average CAM intensity	0.34336814284324646
Representative selection count	25
Generation timestamp	2026-07-30 20:03:14
Dataset version	APTOS 2019 Blindness Detection
Dataset test split checksum	50e87c94f5ba84a457d21d46b27744e58fe05091ce5a7099cdd804c435b04890
Image size	224
Normalization	ImageNet
Checkpoint SHA256	263e8a3f9a9fb7aa995a8aafc40f13b9741d7da02889cf69b3168a9e569f17f2
Git commit hash	a31a2d5711049b778807e846926cd01346cda7fe
XAI method version	1.0.0
CLI Command	src/xai.py --model efficientnet_b3 --checkpoint experiments/efficientnet_b3/checkpoints/best_model.pt --method
Selected representative count	25
Python version	3.12.2 (tags/v3.12.2:6abddd9, Feb 6 2024, 21:26:36) [MSC v.1937 64 bit (AMD64)]
Torch version	2.11.0+cpu
Torchvision version	0.26.0+cpu

CUDA version	N/A
GPU	N/A
OS	nt

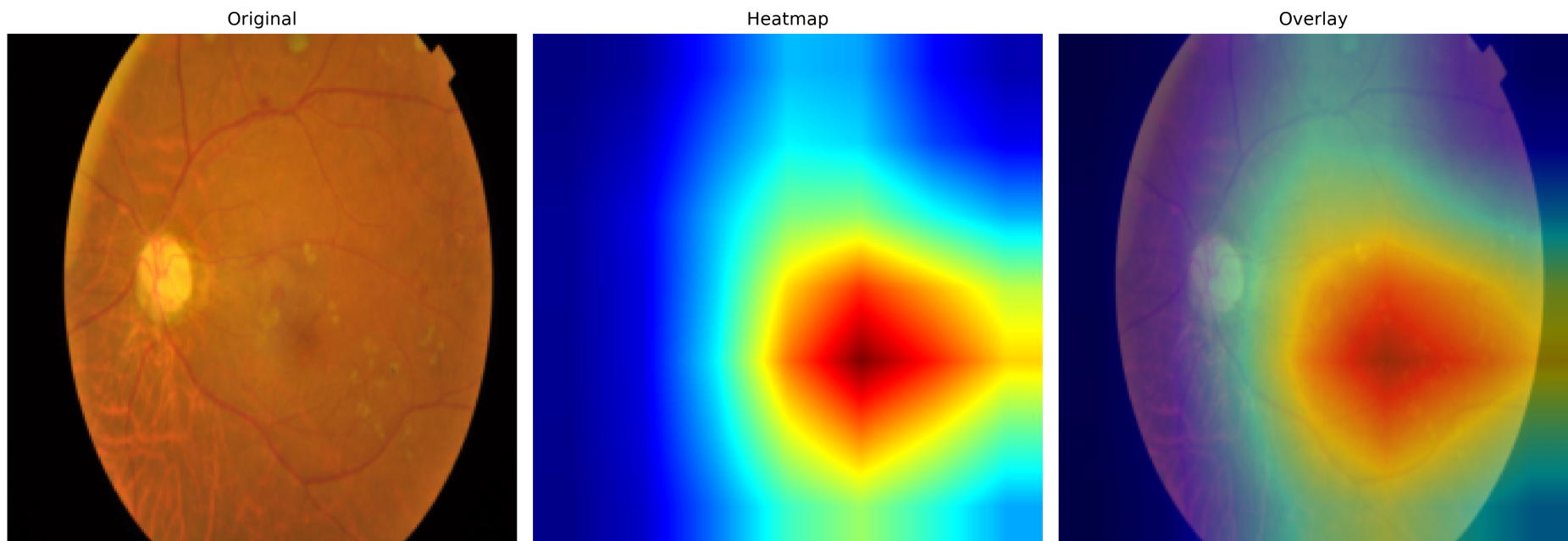
XAI Visualizations Gallery



Ground Truth: Moderate Prediction: Moderate Confidence: 0.9422
Entropy: 0.2361 Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

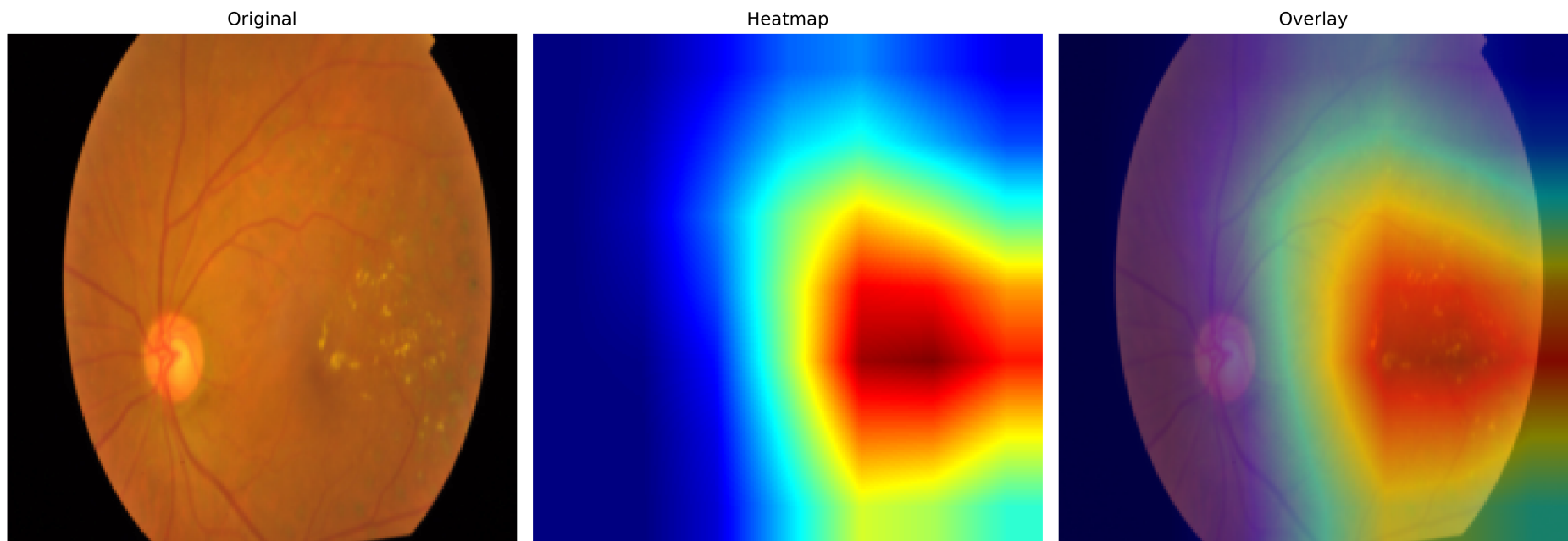
Figure 1: image_10eefba568dd XAI Analysis



Ground Truth: Proliferative Prediction: Moderate Confidence: 0.9793
Entropy: 0.1074 Reason: Severe confusion example (>1 grade error)

Observation: Attention is highly localized to a single region.

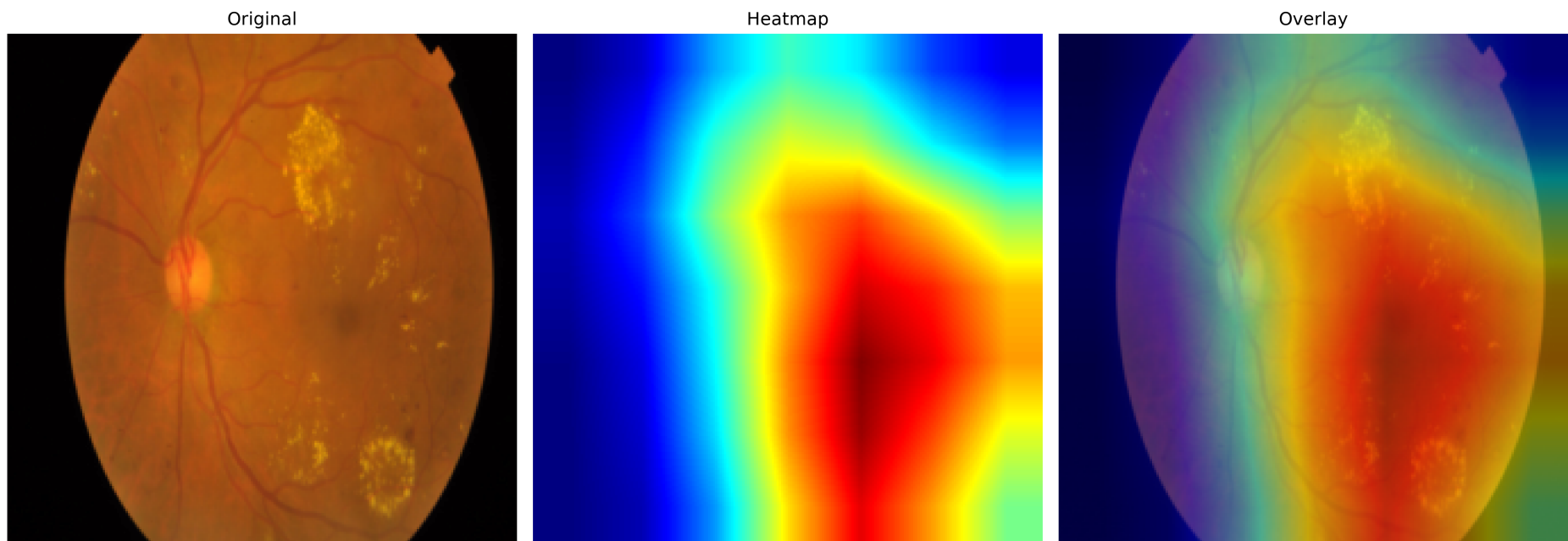
Figure 2: image_1c4f3aa4df06 XAI Analysis



Ground Truth: Moderate Prediction: Moderate Confidence: 0.9998
Entropy: 0.0023 Reason: Representative correct (Grade Moderate)

Observation: Attention is highly localized to a single region.

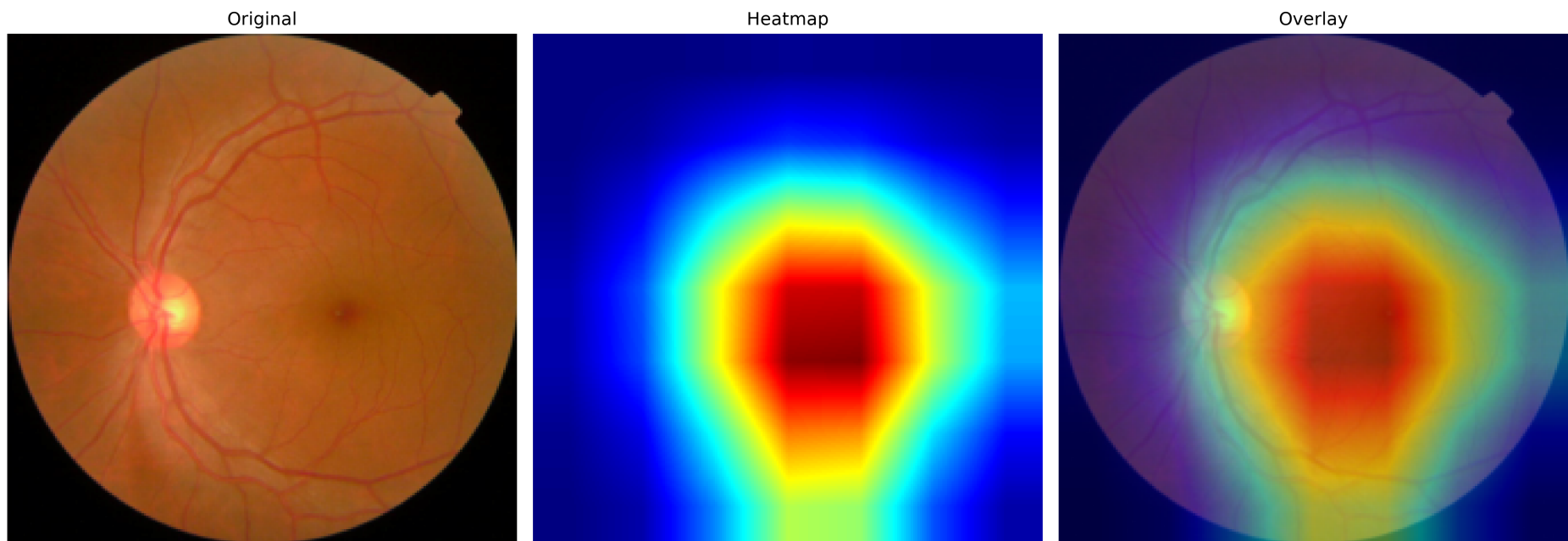
Figure 3: image_23175b7ef453 XAI Analysis



Ground Truth: Severe Prediction: Moderate Confidence: 0.9544
Entropy: 0.2014 Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue.

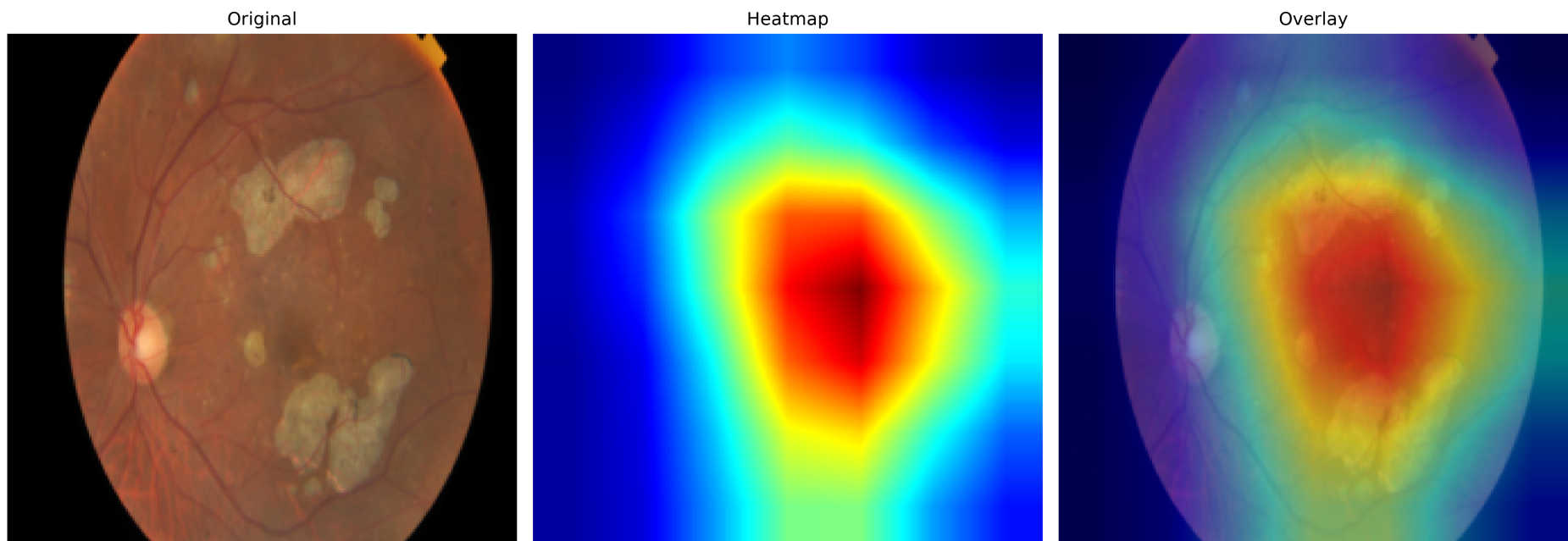
Figure 4: image_24b87f744598 XAI Analysis



Ground Truth: No DR Prediction: No DR Confidence: 0.9997
Entropy: 0.0033 Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

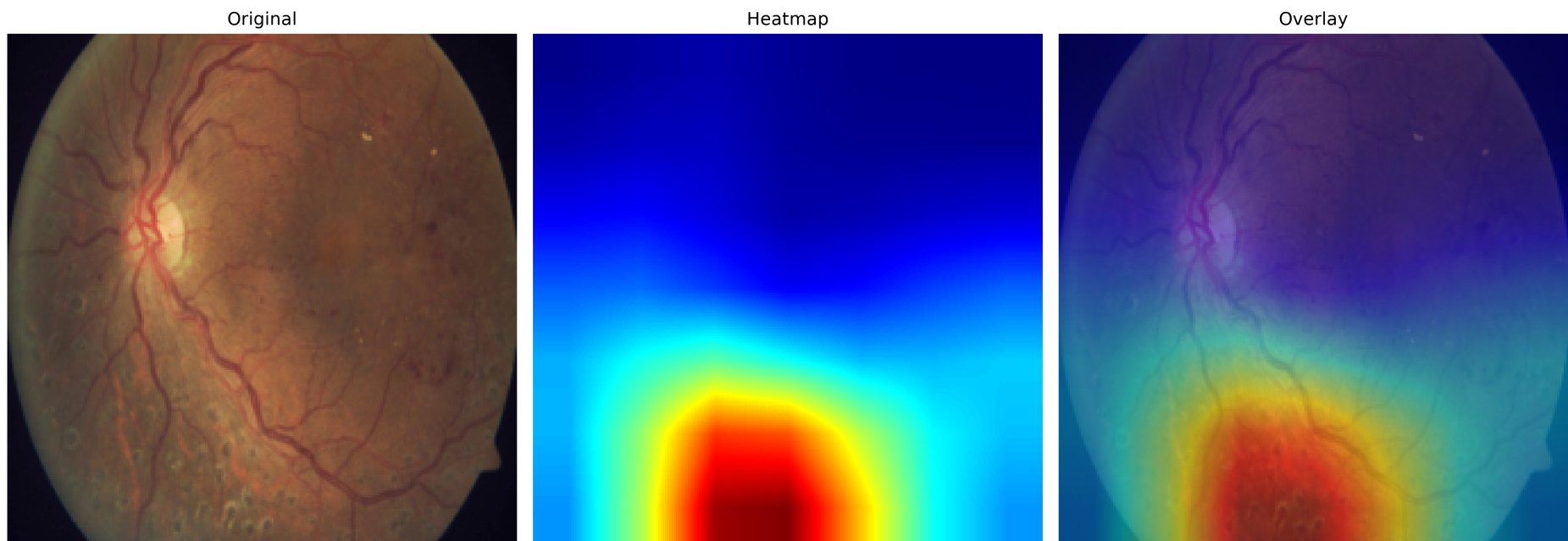
Figure 5: image_2cacdb0dffae XAI Analysis



Ground Truth: Moderate Prediction: Moderate Confidence: 0.5023
Entropy: 0.7650 Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention concentrated near the central (macular) region.

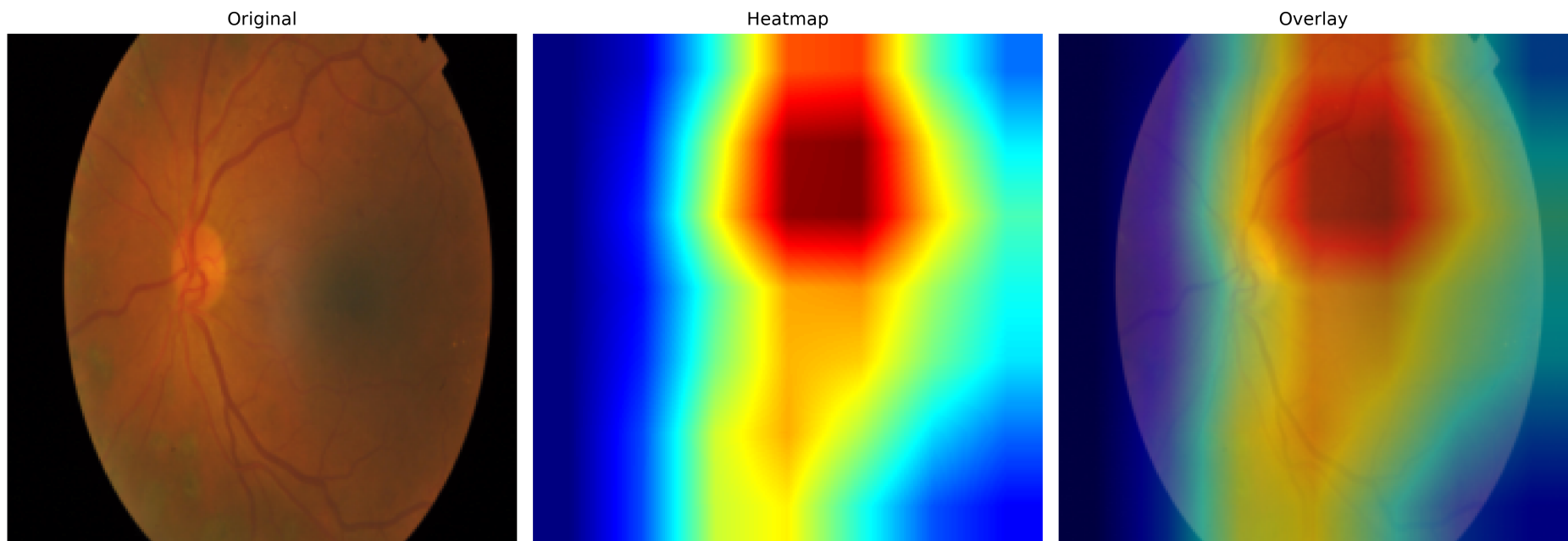
Figure 6: image_2cbfc6182ba2 XAI Analysis



Ground Truth: Severe Prediction: Proliferative Confidence: 0.9977
Entropy: 0.0179 Reason: Most confident incorrect

Observation: Attention is highly localized to a single region.

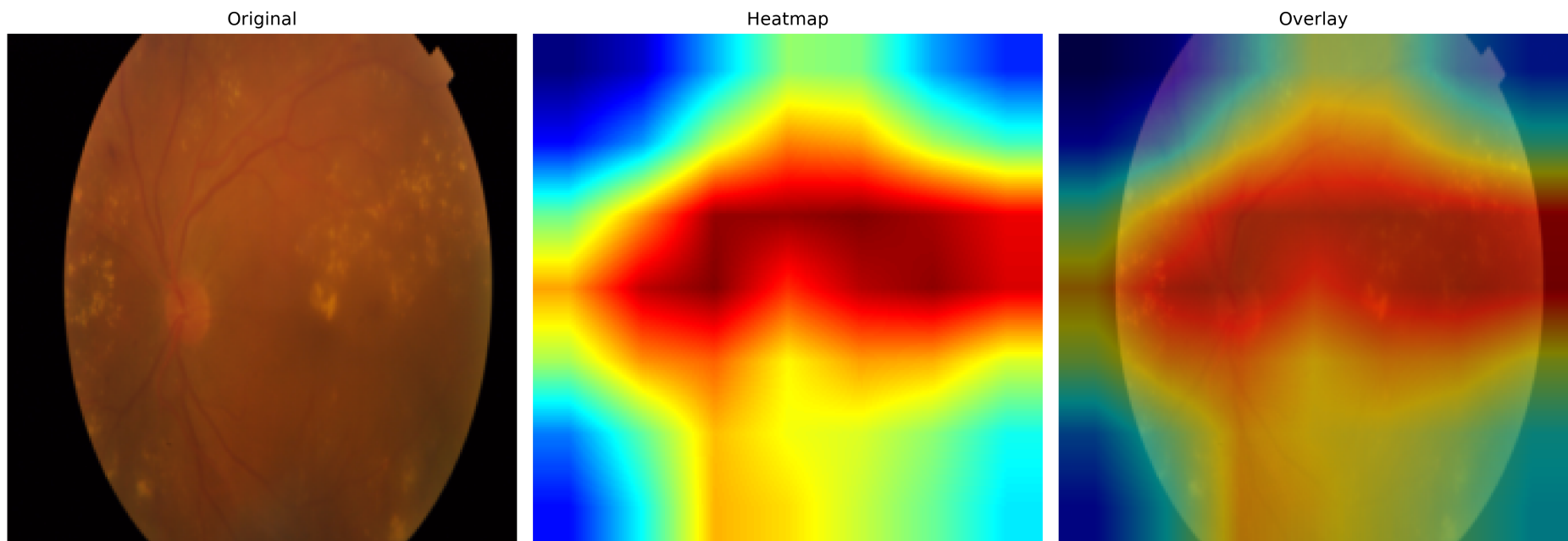
Figure 7: image_3435fd8675a2 XAI Analysis



Ground Truth: Moderate Prediction: Moderate Confidence: 0.5764
Entropy: 0.9036 Reason: Random representative fill

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue. Attention concentrated near the central (macular) region.

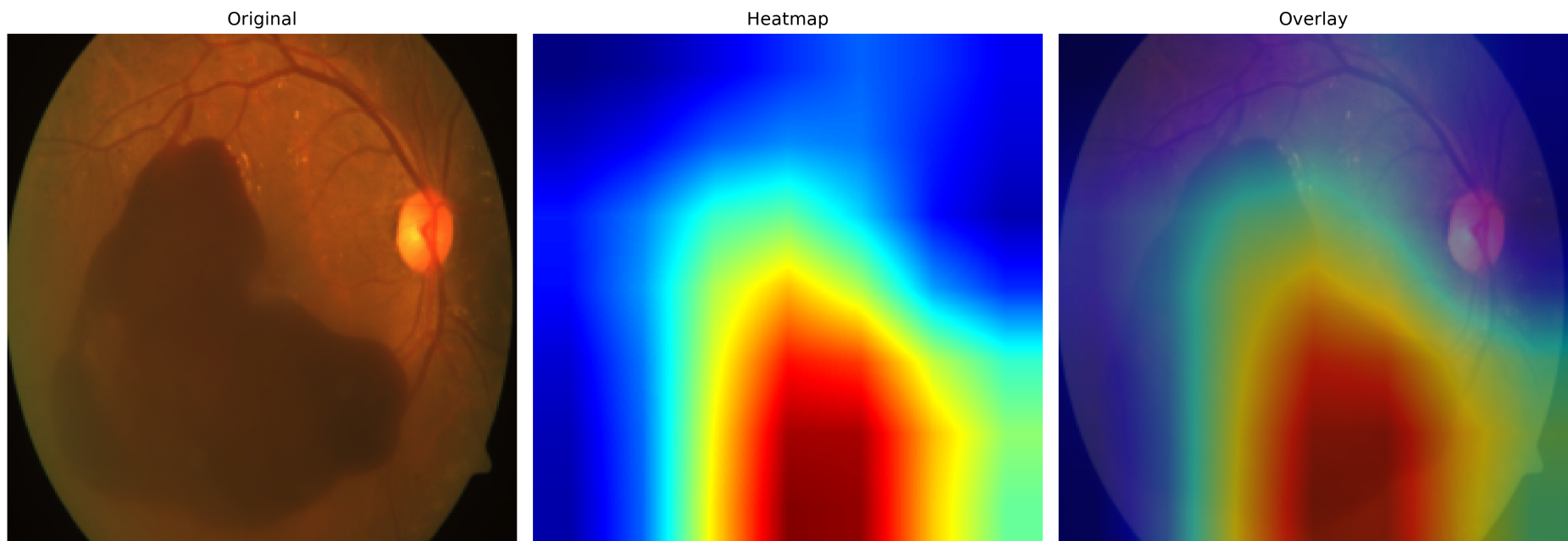
Figure 8: image_453d553b0a94 XAI Analysis



Ground Truth: Moderate Prediction: Severe Confidence: 0.9888
Entropy: 0.0694 Reason: Failure example (Grade Moderate)

Observation: Attention is highly localized to a single region. Attention is widely dispersed across retinal tissue. Attention concentrated near the central (macular) region.

Figure 9: image_4a44cc840ebe XAI Analysis



Ground Truth: Proliferative Prediction: Proliferative Confidence: 0.9988
Entropy: 0.0111 Reason: Representative correct (Grade Proliferative)
Observation: Attention is highly localized to a single region.

Figure 10: image_4bd941611343 XAI Analysis